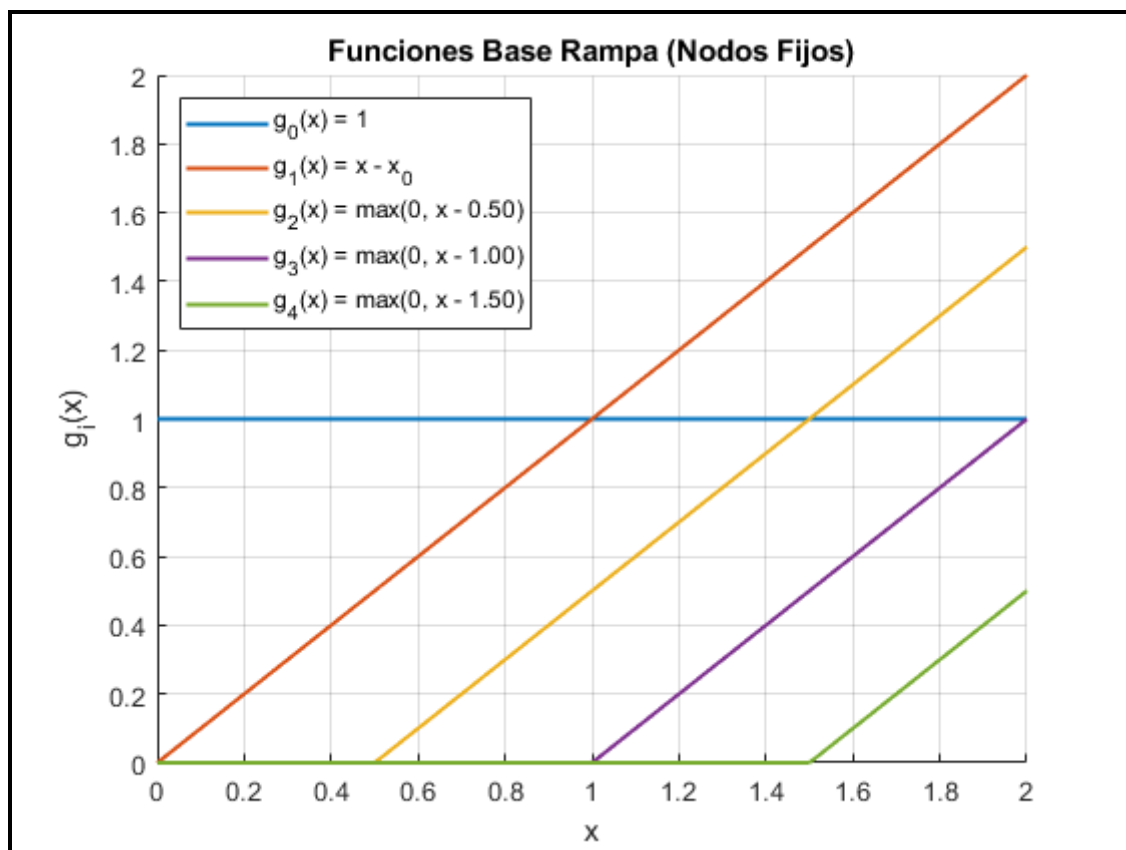
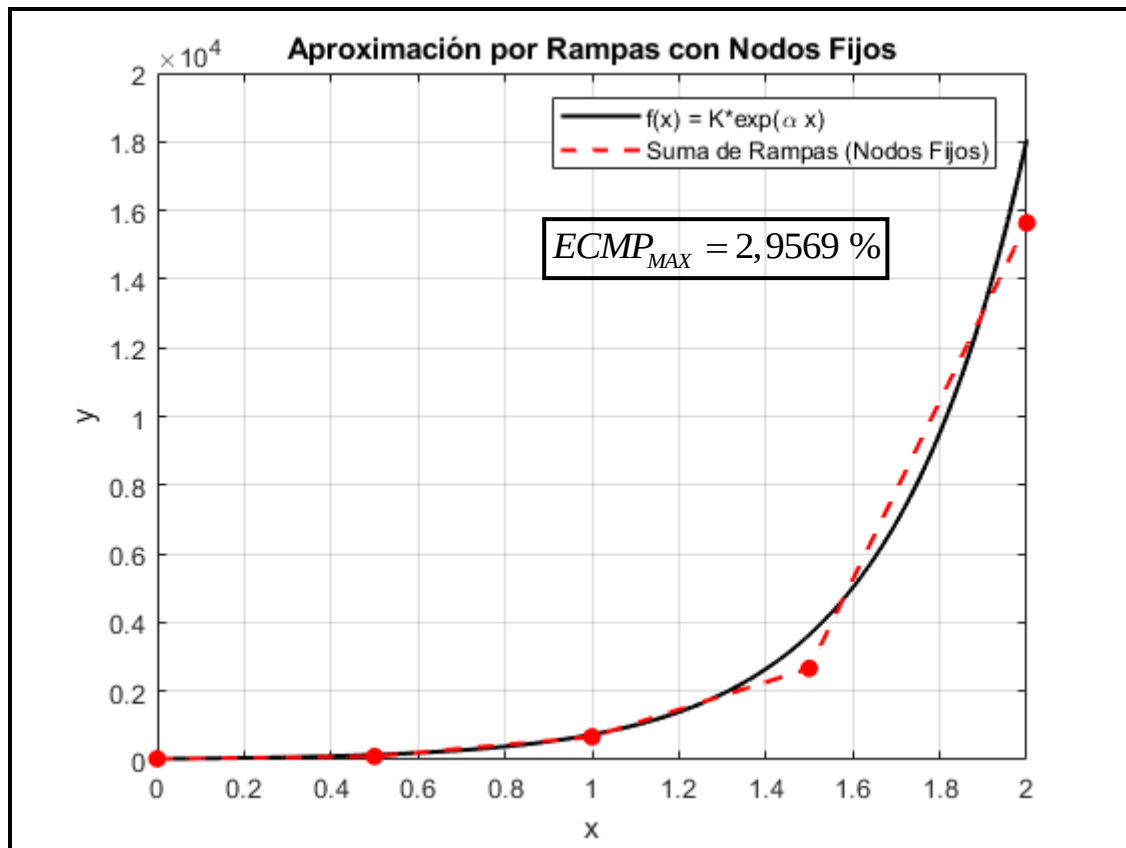


Pruebas de fitting con mínimos cuadrados – Funciones Base tipo Rampa

Mínimos Cuadrados con Nodos fijos – Minimiza el Error L_2 Global



$$\hat{y}(x) = a_0 \cdot g_0(x) + a_1 \cdot g_1(x) + a_2 \cdot g_2(x) + a_3 \cdot g_3(x) + a_4 \cdot g_4(x)$$

$$a_0 = +33,3417$$

$$a_1 = +130,8595$$

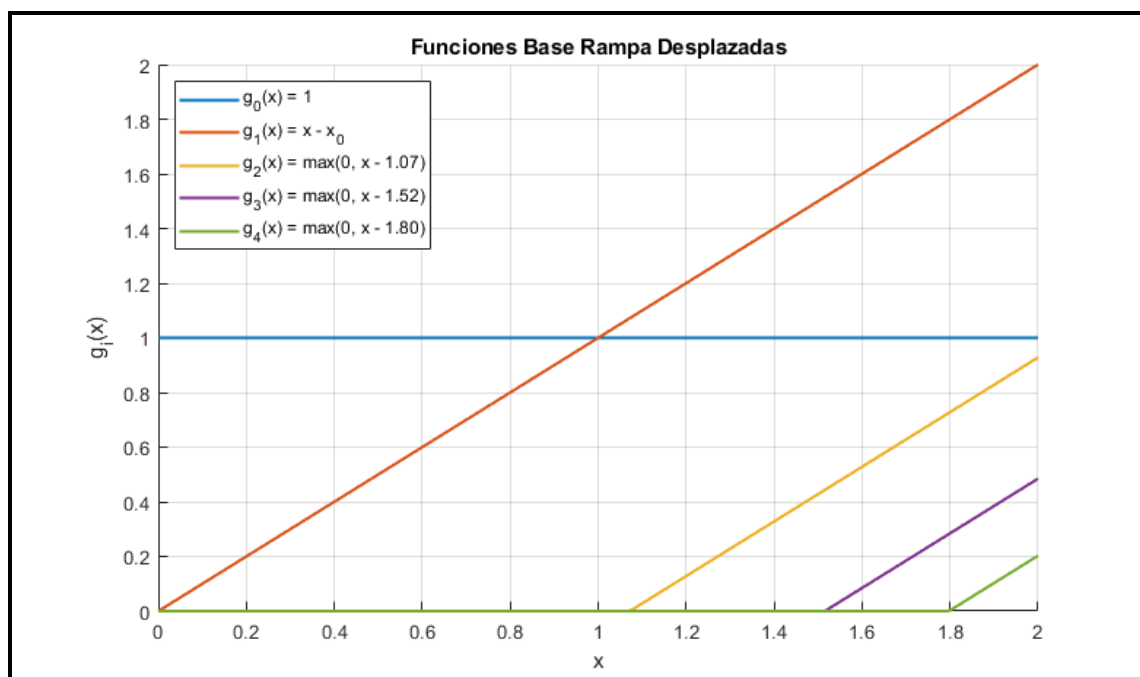
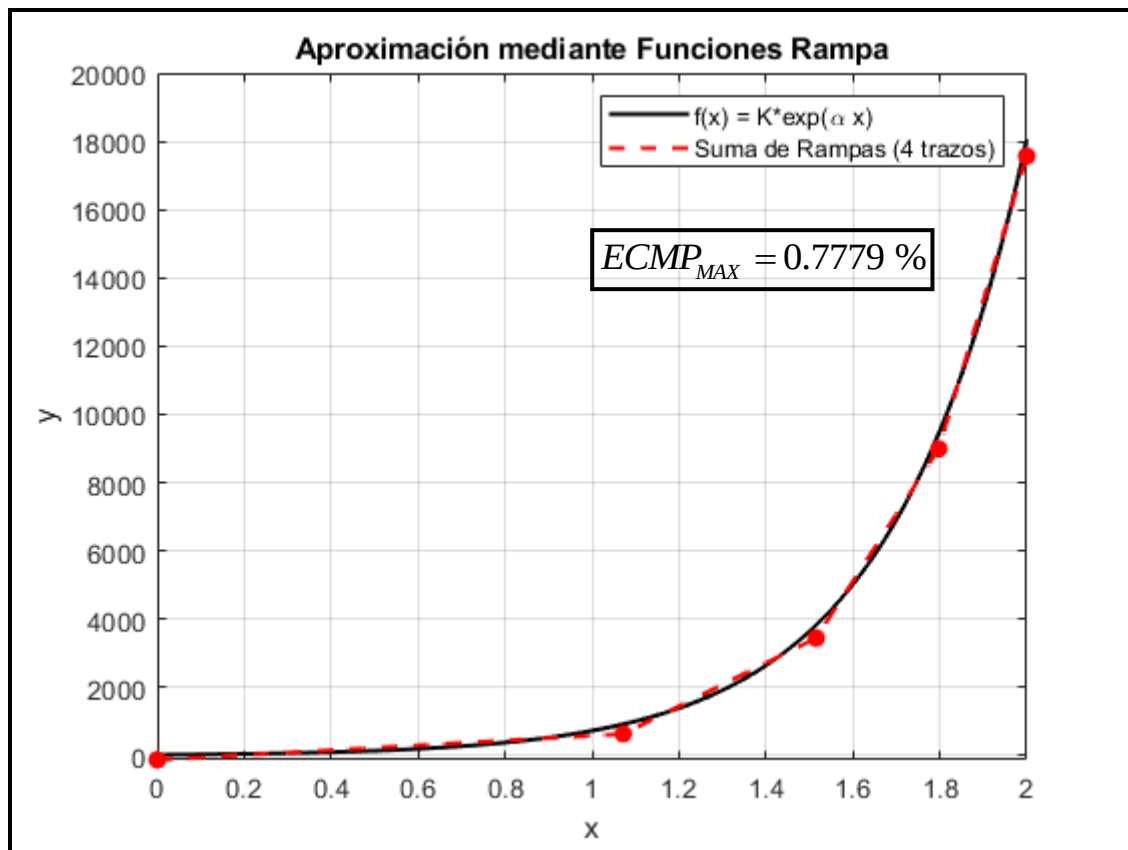
$$a_2 = +1012,1818$$

$$a_3 = +2840,6435$$

$$a_4 = +21963,9429$$

Pruebas de fitting con mínimos cuadrados – Funciones Base tipo Rampa

Optimización Global de Nodos + Mínimos Cuadrados Continuos



$$\hat{y}(x) = a_0 \cdot g_0(x) + a_1 \cdot g_1(x) + a_2 \cdot g_2(x) + a_3 \cdot g_3(x) + a_4 \cdot g_4(x)$$

$$a_0 = -118,1060$$

$$a_1 = +708,0627$$

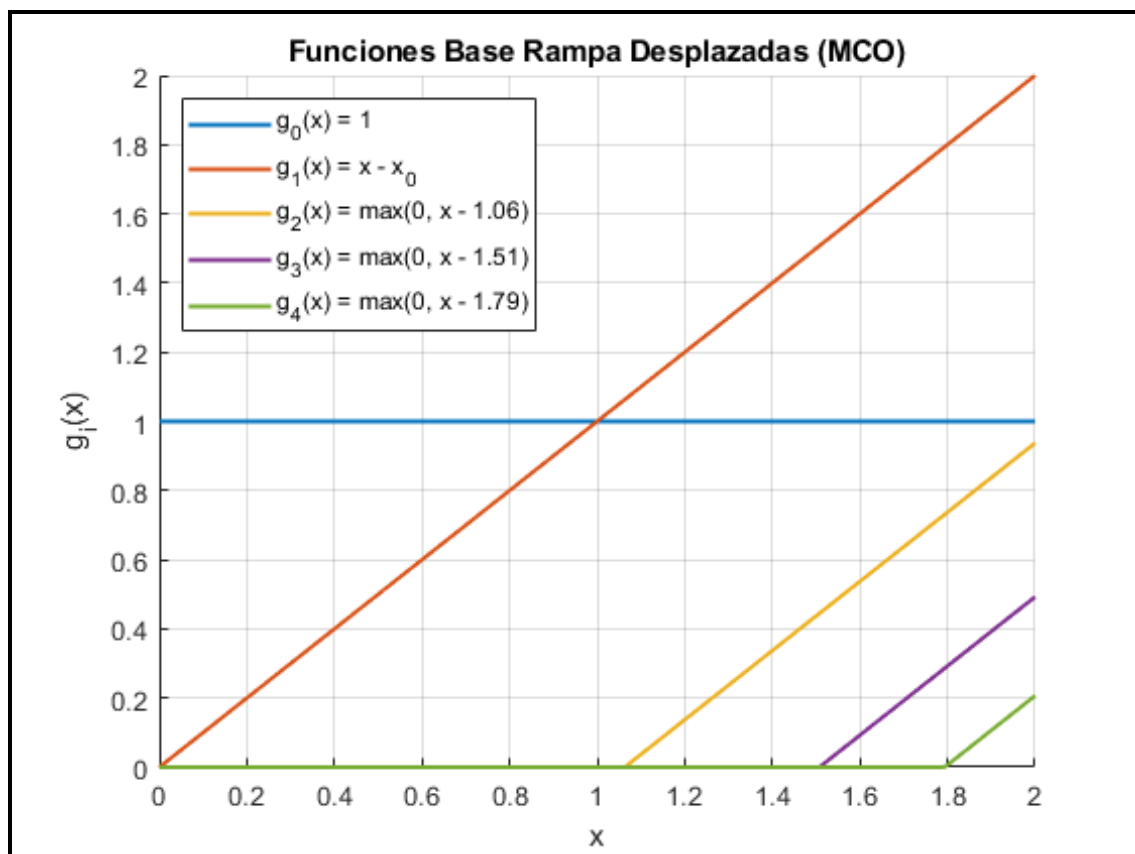
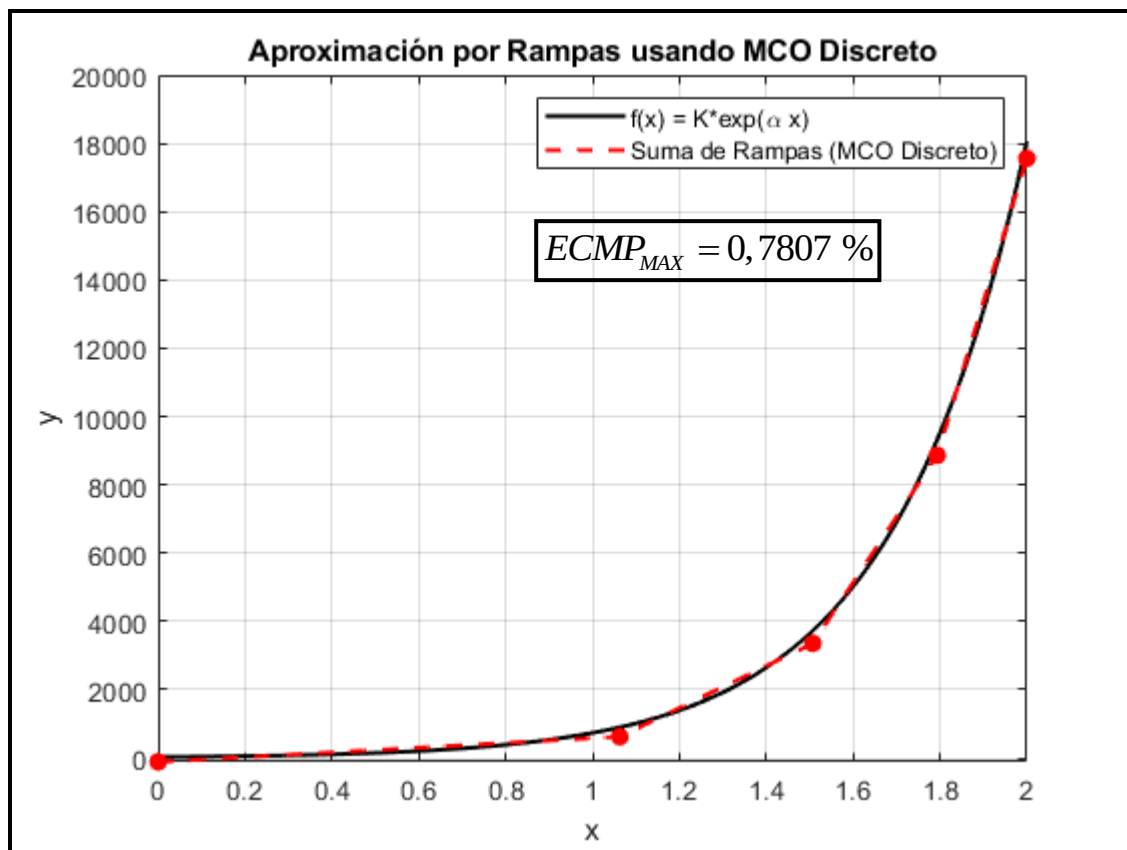
$$a_2 = +5622,0152$$

$$a_3 = +13332,7097$$

$$a_4 = +22649,3532$$

Pruebas de fitting con mínimos cuadrados – Funciones Base tipo Rampa

Optimización Global de Nodos + Mínimos Cuadrados Discretos – MCO



$$\hat{y}(x) = a_0 \cdot g_0(x) + a_1 \cdot g_1(x) + a_2 \cdot g_2(x) + a_3 \cdot g_3(x) + a_4 \cdot g_4(x)$$

$$a_0 = -112,4513$$

$$a_1 = +692,9105$$

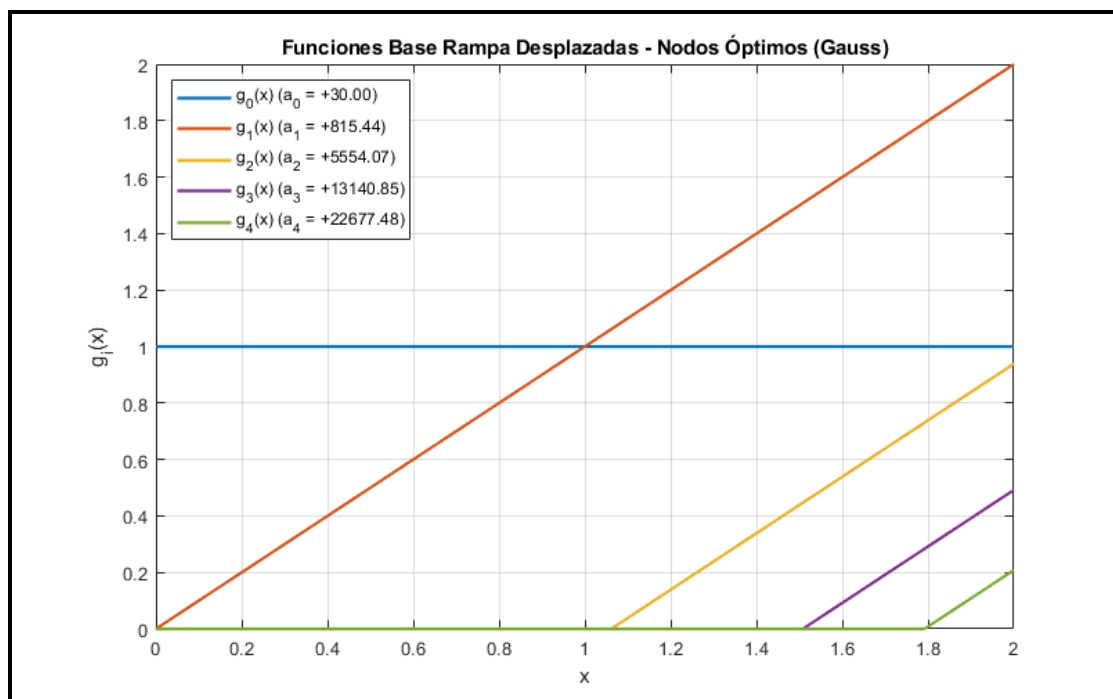
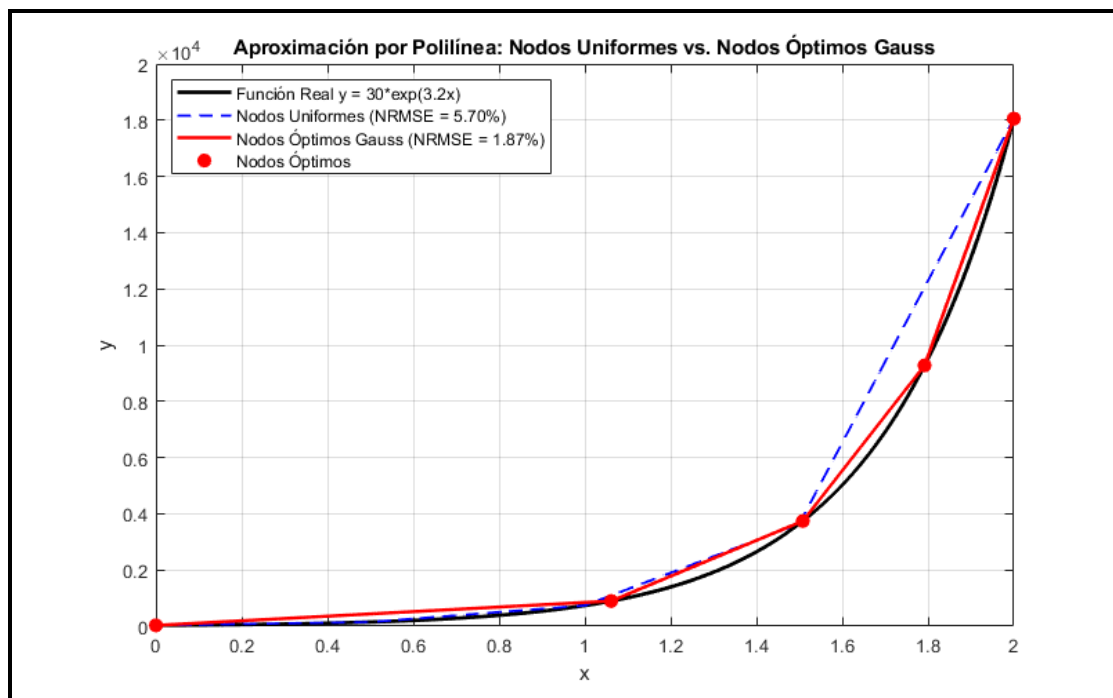
$$a_2 = +5453,1958$$

$$a_3 = +13114,2059$$

$$a_4 = +22831,4480$$

Pruebas de fitting con mínimos cuadrados – Funciones Base tipo Rampa

Nodos uniformes y Nodos Óptimos (via Cuadratura de Gauss)



$$\hat{y}(x) = a_0 \cdot g_0(x) + a_1 \cdot g_1(x) + a_2 \cdot g_2(x) + a_3 \cdot g_3(x) + a_4 \cdot g_4(x)$$

$$a_0 = +30,00$$

$$a_1 = +815,44$$

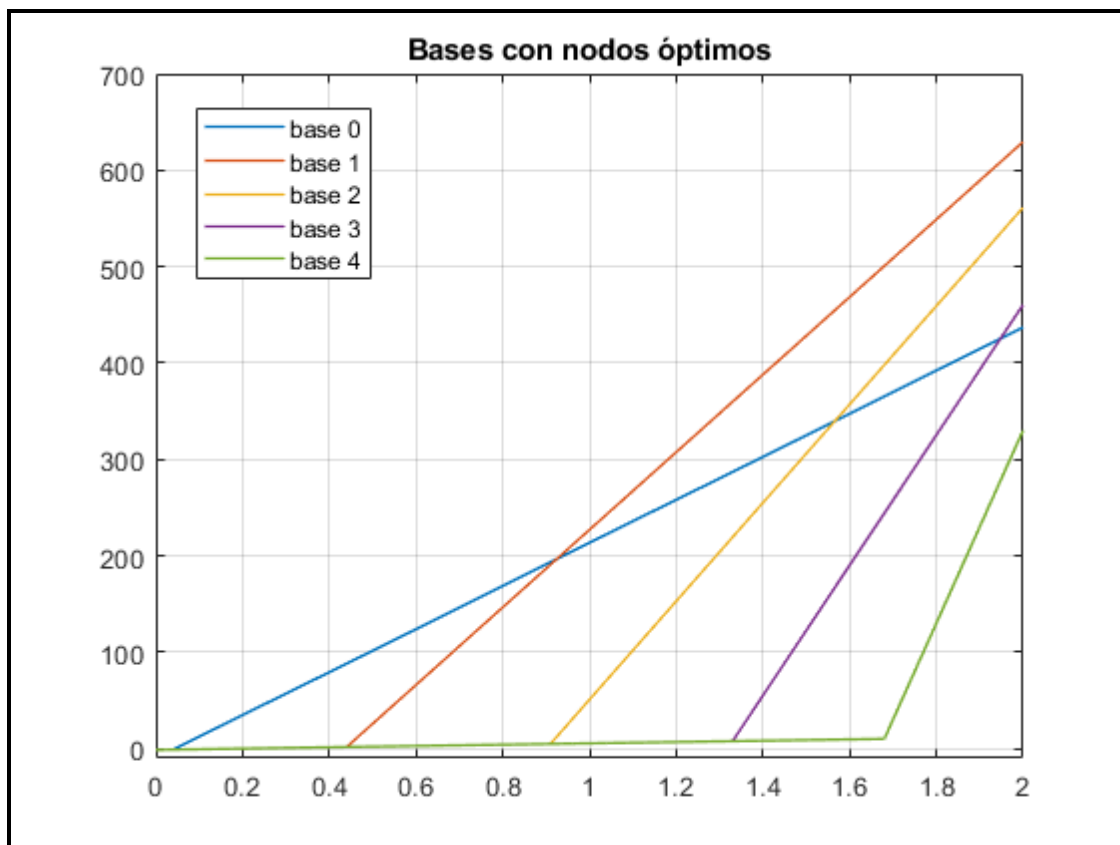
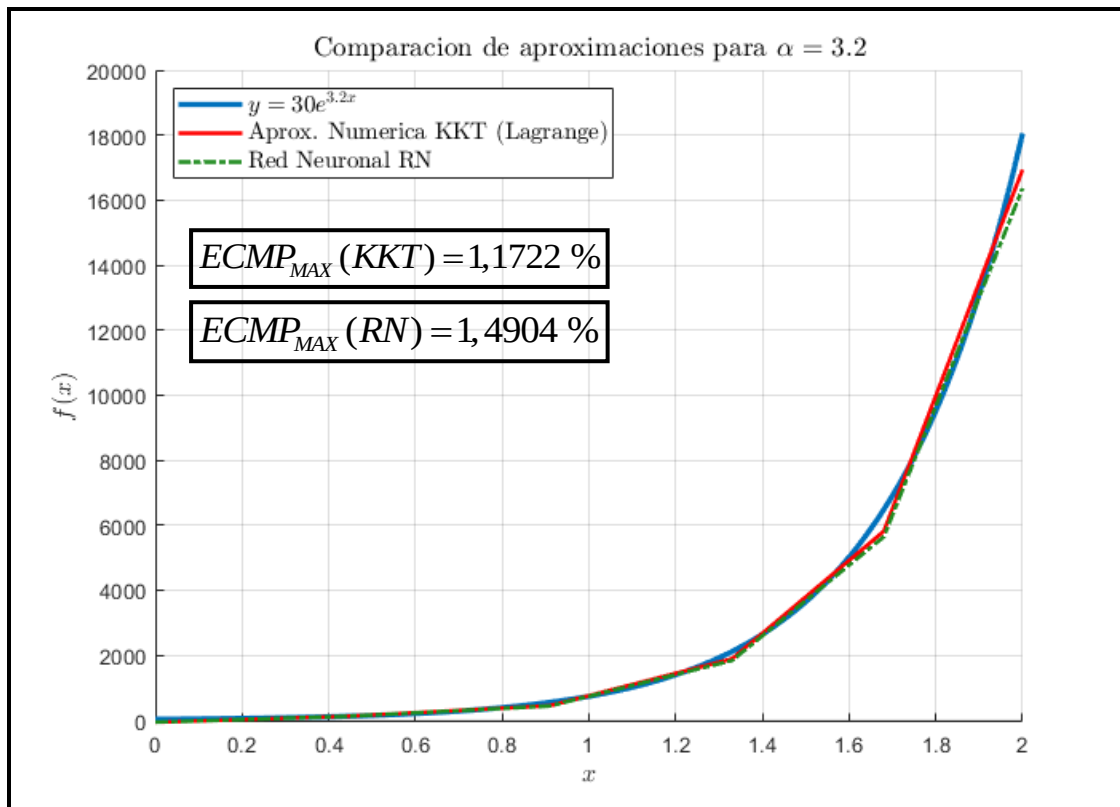
$$a_2 = +5554,07$$

$$a_3 = +13140,85$$

$$a_4 = +22677,48$$

Pruebas c/ Optimo Directo Restricciones KKT – Funciones Base tipo Rampa

Óptimo directo Restricciones KKT y Red Neuronal



$$\hat{y}(x) = a_0 \cdot g_0(x) + a_1 \cdot g_1(x) + a_2 \cdot g_2(x) + a_3 \cdot g_3(x) + a_4 \cdot g_4(x)$$

KKT – Lagrange

$$\begin{aligned}a_0 &= +0,6209 \\a_1 &= +0,6830 \\a_2 &= +5,5433 \\a_3 &= +11,6355 \\a_4 &= +23,7054\end{aligned}$$

RED NEURONAL

$$\begin{aligned}a_0 &= +0,6162 \\a_1 &= +0,6735 \\a_2 &= +5,2671 \\a_3 &= +11,2854 \\a_4 &= +22,7836\end{aligned}$$